

Jacob Dauphinais

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EDUCATION

University of Rhode Island, Kingston, RI

May 2025

Bachelor of Science in Computer Science, Bachelor of Arts in Mathematics, Minor Cyber Security

GPA: 3.62/4.0, Dean's List

Roderick P.C. Caldwell Math Department Scholarship Endowment, Presidential Scholarship

TECHNICAL SKILLS

Programming Languages: Python, Rust, C, C++, JavaScript (Node.js), SQL, Java (Beginner), Kotlin (Beginner), Bash

Web Development: HTML, CSS, Node.js, PHP, React (Beginner), RESTful APIs, Google AdMob, Google Firebase (RTDB, Auth, Hosting)

Tools & Platforms: Git, GitHub, Jupyter Notebooks, MySQL, HPC Clusters, DBMS, Linux/Mac/Windows Systems

Other Skills: MATLAB, Microsoft Office Suite, Google Workspace

PROFESSIONAL EXPERIENCE

CAREERS Cyberteam, Remote

February 2025 - May 2025

Researcher & Developer

- Conducted multimodal analysis of climate change-related social media posts by refining data pipelines and utilizing Large Language Models.
- Utilized Jupyter Notebooks solutions to manage Python solutions and interactions with external data pipelines and High Performance Computing (HPC) environments.
- Deployed multiple LLMs for image captioning on the UNITY HPC cluster to ensure scalable and efficient processing of visual media.
- Integrated the CLIP Text-Image model to select high-quality image captions, improving alignment between textual and visual modalities for downstream analysis.

University of Rhode Island, Department of Nutrition & Food Sciences, Kingston, RI

June 2024 - August 2024

Backend Developer

- Designed and deployed robust APIs in Node.js, ensuring seamless API integration and efficient server-side logic to support application needs.
- Collaborated with cross-functional teams to define, design, and deploy new features, ensuring alignment with project goals.
- Analyzed datasets with SQL to guide feature development, improving data-driven decision-making.

University of Rhode Island, Office of International Education and GSO Campus, Kingston/Narragansett, RI

September 2021 - May 2023

On-site Undergraduate Information Technology Worker

- Provided dedicated on-site service to the University of Rhode Island Office of International Education.
- Created and further developed a Google Drive physical backup system for all digital office files.
- Enforced and provided university standard methods of cyber security measures, such as Multi-Factor Authentication, bitlocker encryption, and administrative software such as Virtual Private Networks or Anti-Virus Software.
- Implemented future plans for purchasing and installing new devices and comparing cost metrics to choose the best option.
- Provided full campus-wide service for the Graduate School of Oceanography directly under the Head of IT at the time.
- Created and implemented approved methods of receiving custom user tickets to allow for a better IT workflow.
- Handled multiple-building device repair, installation, and updating including printers, servers, and user devices.

PROJECTS

Full-stack Web Based Social Media Application

- Collaborated with a cross-functional team of four in planning, designing, and developing a social media application focused on haircuts.
- Implemented the SCRUM framework for effective project management and role designation.
- Utilized GitHub for version control, implementing a branching strategy to facilitate collaborative development and resolve conflicts seamlessly.
- Developed a responsive web app using React, interfacing with a Realtime Database in Google Firebase for efficient data storage and retrieval.
- Successfully integrated Google AdMob for advertisement capabilities, and properly enabled tools for future analysis.

Facial Classification Model Using Olivetti Faces and Labeled Faces in the Wild

- Redesigned and implemented a pre-existing facial recognition/classification algorithm to test on different facial recognition datasets.
- Utilized Python's Matplotlib to visually display data such as the faces, centroids, eigenfaces, and other pertinent information.
- Documented shortcomings of the pre-existing algorithm, and the importance of well cleaned and standardized datasets for image and especially facial recognition models.
- Applied k-fold cross-validation to evaluate the model's performance more robustly across different datasets, ensuring more reliable accuracy metrics and minimizing overfitting.